Sample & Population - Practice Excercises

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Recommended Matlab: >= 2023a.

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Usage Rights

These materials were created for educational purposes and to be completed as part of material covered in the CE305: Probability, Statistics, and Data Analysis undergraduate course. Students will solve this practice notebook as part of the weekly assignment and turn it to be graded for completion and not correctness towards the participation category of their final grade.

Learning Objectives

By the end of this notebook, students will be able to:

- Define and identify sample and population;
- Utilize descriptive statistics to compute metrics for sample data;
- Identify sample space and its events
- Apply axioms of probablity to perform operations between events;
- Recognize the nature of experiments and use the appropriate counting method to determine expected number of outcomes.
- Compute probabilities of events for a known sample space.

Practice Activities for Notebook 1

Submit this notebook along with your problem set assigned for Week 2 (due September 5th, 2025)

Activity 1: True or False

1. We want to understand the average study duration of all students in the school, so we survey one Grade 12 class. These surveyed students make up the population.

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2. If we use a random number table to select 50 students from the entire school registry, this constitutes a simple random sample.

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3. A sample must be larger than the population in order to produce accurate conclusions.

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4. The population refers to the entire group of interest, while a sample refers to the subset of data used to study trends.

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5. Surveying your 10 closest friends on their favorite course this semester is a simple random sampling method.

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Activity 2: Statistical analysis of napping times

You are part of your school's *"Healthy Detective"* project. Your mission is to investigate whether taking naps during the day helps improve students' learning performance. You randomly select 10 students and record their average nap duration per day (in hours) over the past week. Here is the data you collected:

Nap Duration Data (hours): 1.0, 0.5, 1.5, 2.0, 1.0, 0.0, 0.5, 3.0, 2.5, 1.5

- Calculate the mean nap duration.

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- Determine the median nap duration.

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- Calculate the sample variance of nap duration.

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- Calculate the sample standard deviation of nap duration.

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Activity 3: Operations with Events

We have a prize box with the following items:

Number	Color	Prize Type
1	Red	Toy
2	Red	Candy
3	Blue	Toy
4	Blue	Toy
5	Yellow	Candy
6	Blue	Candy
7	Red	Toy
8	Yellow	Toy
9	Blue	Candy
10	Red	Candy

Questions:

- What is the size of sample space S?

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- Given event A: "Drawing a red prize." How many elements does this event have?

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- Given event B: "Getting a candy prize." How many elements does this event have?

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- What is the probability P(A ∪ B)?

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- What is the probability P(A ∩ B)?

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- What is the probability P(B^C)?

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- Are there any mutually exclusive events? If yes, please list; if no, why?

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Activity 4: Counting

Given the following tasks:

- Task A requires two people,
- Task B requires one person,
- Task C also requires one person.

You select four people out of ten to complete each a single task (no person completes more than one task).

Question:

How many different ways are there to assign these four people to complete the tasks?

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